

Research Proposal

From Proximal Gains to Global Savings: Unifying Data Valuation across Text and Trajectory Data

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1. Abstract

Current LLM pre-training pipelines treat training samples as interchangeable, independent units, and virtually all existing data-selection and scoring tools rest on this exchangeability assumption. The assumption fails precisely where it matters most: training is non-convergent and multi-stage, data arrive sequentially, sample value drifts as training proceeds, and—in embodied intelligence—data take the form of temporally dependent trajectories, the costliest modality to collect. This project bridges the methodological divide between data selection for LLM pre-training and for trajectory-based embodied training, establishing a single framework of data refinement and data scoring—resting on a shared mathematical core—that covers both data regimes. The research unfolds in three thrusts. **(RQ1) Failure boundaries, constructively.** We systematically chart when and how badly proximal data-valuation tools fail under a threefold deviation—dataset-level granularity, sequential value drift, and distribution shift—decomposing their error into four correctable sources (truncated horizon, missing conditioning, broken exchangeability, and selection-feedback shift), and characterizing when per-sample independent scoring ceases to be well-defined. **(RQ2) Prospective valuation.** We construct a history-conditioned value function $V(D \mid H_t, B)$ —defined on model parameters, optimizer internals, and remaining training budget—together with a training-free estimator obtained not as a new primitive but by repairing the proximal backbone term by term: one dedicated correction module per error source isolated in RQ1, composed in a canonical causal order, with the residual interaction among corrections explicitly measured rather than assumed away. Thus, RQ1’s decomposition is the design blueprint of RQ2’s estimator, and RQ2’s estimator is the constructive validation of RQ1’s diagnosis. **(RQ3) Generalization to trajectories.** We extend the framework from exchangeable samples to temporally dependent trajectory data for vision-language-action policies, delivering a trajectory-aware scorer (segment-level causal correction, trajectory-level group influence, distribution-level shift gating), training-time stopping criteria, and a cross-modal benchmark showing that one frozen value function covers text and trajectories alike. Ultimate goal: for LLMs and embodied decision models, given the current model state, optimizer internals, and remaining budget, estimate—without any actual training—the expected capability gain of admitting a candidate dataset, thereby preserving downstream performance while reducing global training consumption in data, tokens, and compute.

2. Research Fit with SPHERE Lab

1. The project sits at the intersection of the group’s two main research lines: efficient and stable training of foundation models, and data-centric modeling of inductive biases (selection / distillation / curriculum). The research answers training-side efficiency with data-side principles: every claim of reduced compute, token, or data consumption is grounded in how data value is defined, estimated, and operationalized. The work therefore speaks to both lines simultaneously rather than being located within either one alone.
2. The core methods have direct lineage within the group. The RQ2 scorer is a direct extension of the CHIPS paradigm [4] (CVPR 2026)—curvature-aware influence estimation with proven correlation lower bounds—from the single-shot to the sequential setting, and from image–text pairs to dataset granularity. Moreover, the optimizer geometry of the value function adopts Pion [105], the group’s spectrum-preserving optimizer: the estimator’s geometric assumptions and the trainer’s actual dynamics are then drawn from one and the same family, eliminating at the source the very mismatch between scoring geometry and training geometry that RQ1 identifies as a principal error term.
3. The embodied deployment of RQ3 has in-group precedent. The group’s LEGO project (2026) has completed the first practice of sample-efficient post-training on physical-reasoning data. The RQ3 extension to trajectory data therefore builds directly upon established in-group infrastructure and expertise—an increment on existing work, not a capability constructed from zero.
4. Methodological taste and theoretical tooling are aligned on both sides of the proposal. The formulation is assembled entirely from field-standard primitives—influence functions, preconditioned updates, scaling laws—with every tool carrying a provable guarantee, in direct execution of the stated credo of “light-yet-sweet: insight must precede application.” The optimizer state that the value function must encode is, in turn, underwritten by the group’s Muon spectral-analysis line [104] (2026), a theoretical tool line unique to the group—ensuring that the most technically demanding component of the proposal rests on the group’s deepest comparative advantage.

3. Scope and Literature Review

3.1 Problems in Focus

The main question of this research: for an LLM or an embodied decision-making model, given the model parameters, the internal state of the optimizer, and the remaining training budget, is it possible—without any actual training, using only lightweight representations extractable from the model’s internals—to estimate whether admitting a given dataset, relative to other data (or to admitting nothing at all), can reduce the model’s global training consumption (data, tokens, compute) while preserving downstream performance?

This overarching question decomposes into three core research points:

- **RQ1:** Probing the failure boundaries of existing proximal methods. In existing research, online importance sampling [97] dynamically reweighted samples throughout training as early as the small-model era, and diversity-aware importance resampling [19] remains the default entry point for pre-training data curation; curriculum learning [20] and dynamic token selection [3] likewise exploit

the fact that order matters. Yet the reality is that these tools have been validated only within their respective settings (weighting within a fixed data pool, token selection within a single corpus, proximal improvement on a fixed proxy set), and their applicability boundaries have never been systematically charted—even the latest critical examination of targeted instruction selection [118] stops at identifying which factors actually matter, leaving unanswered the conditions and magnitude of failure. Specifically, existing work does not address three questions. First, when the object of scoring changes from a single sample to an entire dataset that is distributionally shifted from the data already seen, does a utility definition aimed at proximal improvement still correspond to long-term expected gains [4][85]? Second, when data value decays or migrates as training proceeds (diminishing returns on repeated reuse under data-constrained regimes [85]; order sensitivity of curricula and anti-curricula [11]), when exactly do proximal scoring mechanisms lose accuracy? Third, when samples are not exchangeable (causal dependence in trajectory data [13])—or when the act of selection itself alters the distribution of subsequent data [49]—is the very operation of per-sample independent scoring still well-defined? The failure boundaries of existing proximal data-valuation methods therefore need to be unified as a single error model whose terms are individually correctable which can serve simultaneously as a critique of existing tools and as the specification of their repair.

- **RQ2:** Constructing data-valuation tools conditioned on training history and aimed at saving global training cost. Widely used data-valuation tools fall into three categories. Model-state-agnostic static scoring. Methods exemplified by quality classifiers (e.g., FineWeb-Edu [24]) grade data before training begins, and once issued the score never changes as training proceeds. Its advantage is low cost; its limitation is equally evident: the same data may differ entirely in value between the early and late stages of training, a fact that static scores cannot reflect. Proximal-horizon targeted scoring. LESS [26] pioneered the paradigm of "selecting influential data for a specific target task"; its successors (e.g., GIST [30]) further show that approximating the update geometry with optimizer diagonal statistics must be replaced by subspace alignment in parameter-efficient fine-tuning settings such as LoRA to avoid failure, and NICE [21], targeting non-differentiable evaluation metrics, extends the same lineage through bi-level optimization. These approaches are coupled to model state, but their definitions rest on next-step or short-term parameter updates rather than on "the full cost from the current point to the target," and they depend on retrospective valuation that requires actual training. Retrospective valuation. Methods such as Data Shapley [6] and Datamodels [7] yield high-precision value estimates, but at the cost of extensive real training (fitting or retraining), and their semantics are "past contribution" rather than "future savings"—rendering them unable to support mid-training decisions. The construction path is fixed by a diagnosis-prescription isomorphism: for each error source isolated in RQ1, the estimator carries one dedicated correction module—truncated horizon is repaired by segment-wise unrolling along the remaining training trajectory ; missing conditioning by projecting utility into the optimizer’s actual update geometry; broken exchangeability by relation-aware group aggregation; selection feedback by a distribution-tracking term that prices the shift the selection act itself induces.
- **RQ3:** Extending data valuation from exchangeable samples to trajectory data. Trajectory data differs from text or image samples in that its value is not attributable to independently exchangeable units. A trajectory contains temporally dependent state-action sequences, and its learning value depends jointly on trajectory-level outcomes, segment coherence, the current policy, the surrounding

data mixture, and embodiment-specific factors [13][36][37][113][119]. Consequently, the value of an individual state–action pair may be ill-defined, while the value of a trajectory or segment may change as training progresses or as the data mixture changes. Existing trajectory-valuation methods largely focus on isolated trajectories or post-hoc influence, leaving these interactions and training-stage dependence insufficiently modeled [37][109].

RQ3 therefore does not introduce a separate valuation paradigm, but evaluates whether the training-history-conditioned framework developed in RQ2 remains valid when the data unit becomes a structured trajectory. Specifically, the framework will be evaluated along three dimensions: whether trajectory-level aggregation can replace independent sample scoring under temporal dependence; whether relation- and mixture-aware corrections can capture interactions among trajectories; and whether the resulting valuation can predict the reduction in training data or compute required to reach a fixed downstream policy-performance target. Where the assumptions of RQ2 fail, the corresponding correction modules will be adapted to trajectory-level temporal, relational, and embodiment structure. This establishes whether a common mathematical core can cover both exchangeable samples and structured trajectory data.

3.2 Problem Background and Clarification of Definitions

- Scoring granularity is the dataset: quality is judged and keep/drop decisions are made at the dataset level.
- Given and invariant: model parameters, optimizer state, and remaining budget are all assumed observable.
- Data exchangeability: a sample sequence $(z_1, \dots, z_n)$ is exchangeable $\iff$ for any permutation σ ,

$$P(z_1, \dots, z_n) = P(z_{\sigma(1)}, \dots, z_{\sigma(n)}).$$

i.i.d. implies exchangeability; the converse fails. Text pipelines are approximately exchangeable at document granularity, so shuffling is harmless; after trajectory segmentation, (s_k, a_k) are treated as “samples,” but the state continuity of

$$P(s_{k+w} \mid s_k, a_k)$$

changes the joint distribution under permutation—non-exchangeable. The marginal value of segment k therefore depends on the other segments of the same trajectory, and per-segment independent scoring omits cross-segment coupling by definition.

- Prospective, not retrospective: in building data-refinement methods, this research is concerned only with potential datasets that have not yet entered training; for data scoring, it assesses only the effectiveness of data already trained on—no retraining from scratch.
- Training-free: data quality is judged via observable indicators without additional training, rather than via the parameters of a fully trained model, so as to reduce training cost.
- Criterion for effectiveness in LLM pre-training: at minimum, “reduced consumption at equal performance” or “improved performance at equal consumption.”
- Explicit exclusions: offline reuse value / OPE [18] and cross-embodiment transfer-volume prediction are not objects of contribution.

3.3 Literature Review

This literature review organizes existing work along two axes: the **data-form axis**, from exchangeable samples to sequentially dependent trajectories, and the **decision-horizon axis**, from proximal improvement to the global cost of reaching a target, with training-freeness and sequentiality as auxiliary attributes. Existing surveys largely cover language-model data selection but do not cross these two dimensions [116], while dataset cartography diagnoses training dynamics without making selection decisions [12].

Existing methods occupy partial regions of this space. Online selection [2][3] is training-free and sequential but proximal; targeted curation and CHIPS [4][26] condition on model state but remain single-shot and short-horizon; CUPID [37] extends valuation to trajectories but retains segment-level independent evaluation; retrospective valuation [6][7] requires actual training, while mixture fitting [8][36] addresses global outcomes but sacrifices training-freeness. Related approaches such as distillation, synthesis, and active learning [27–29] address data compression or acquisition rather than future training-cost savings.

The resulting gap is **training-free, history-conditioned, sequential valuation of dataset-level additions across both exchangeable and trajectory data**. RQ1 characterizes the failure of existing proximal methods as the horizon expands; RQ2 constructs the corresponding history-conditioned value function; RQ3 extends and evaluates it on sequentially dependent trajectory data.

4. Problem Formulation

- LLM model state:

$$H_t = (\theta_t, P_t, D_{\text{seen}}, b_{\text{used}})$$

- LLM target performance

Empirical risk minimization (ERM):

$$L(\theta; D) = \frac{1}{n} \sum_{i=1}^n \ell(f_\theta(x_i), y_i), \quad \theta^* = \arg \min_{\theta} L(\theta; D)$$

- Optimizer state P_t

$$\theta_{t+1} = \theta_t - \eta_t P_t \nabla_{\theta} L(\theta_t; B_t)$$

(B_t is the mini-batch at step t , and P_t is the optimizer’s preconditioning matrix; the optimizer may be SGD, AdamW, Muon, or Pion [57][59][104][105].)

- Proximal quality of a single data point (proximal utility)

$$U_{\text{prox}}(z; \theta_t) = \langle P_t \nabla_{\theta} \ell(z; \theta_t), g_{\text{proxy}}(t) \rangle - \lambda R(z)$$

(R denotes a redundancy term; λ is a coefficient.)

- Trajectory:

$$\tau = (s_0, a_0, s_1, \dots, s_H)$$

- The optimal policy satisfies the Bellman equation:

$$\mathcal{L}_{\text{BC}}(\theta) = \mathbb{E}_{\tau \sim D} \left[\frac{1}{H} \sum_k \ell(f_\theta(s_k), a_k) \right]$$

- The optimal policy satisfies the Bellman equation:

$$V^*(s_t) = \max_a [r(s_t, a) + \gamma V^*(s_{t+a})]$$

- Stopping criterion:

$$\tau^* = \inf \{t : \mathbb{E}[J(\theta_{t+1}) - J(\theta_t)] \leq \lambda \cdot c_{\text{step}}\}$$

- Equivalent cost-saving form:

$$S(D \mid H_t) = C_{\emptyset}(\text{target}) - C_D(\text{target}), \quad \text{where } C = \text{Cost}.$$

- Given the training history H_t , the remaining budget B , the future training algorithm A (including its data-selection policy), and a measurable capability metric $J(\theta)$:

$$V(D \mid H_t, B) = \mathbb{E}_A [J(\theta_T^D)] - \mathbb{E}_A [J(\theta_T^{\emptyset})]$$

where θ_T^D denotes the parameters reached at the termination time when D is introduced during the remaining training, and $\theta_T^{\emptyset}$ is the counterfactual control in which D is not introduced.

- Error model and constructor specification. For any proximal scorer f in the family under test, let $E(f)$ denote its error against the prospective value V under the threefold deviation. RQ1 hypothesizes and measures the decomposition

$$E(f) \approx \varepsilon_{\text{horizon}} + \varepsilon_{\text{cond}} + \varepsilon_{\text{exch}} + \varepsilon_{\text{fb}} + \varepsilon_{\text{int}},$$

where the first four terms are defined operationally by module-based completion (E1.2) and ε_{int} collects residual interactions among them. The RQ2 estimator is then specified:

$$\hat{V} = f_{\text{prox}} + \Delta_{\text{cond}} + \Delta_{\text{exch}} + \Delta_{\text{horizon}} + \Delta_{\text{fb}},$$

with one Δ per ε : (i) Δ_{cond} —geometry correction is a per-step property of the optimizer, independent of aggregation or horizon; (ii) Δ_{exch} —group aggregation operates on geometry-corrected per-unit scores; (iii) Δ_{horizon} —unrolling propagates corrected aggregates across the remaining training trajectory; (iv) Δ_{fb} —feedback correction is self-referential, pricing the shift induced by the act of selecting, which presupposes the first three corrections. Each stage consumes the outputs of the previous one.

5. Related Work and State of the Field: Methods, Progress and Limits

5.1 Data-Scoring Methods for Datasets Already Formed

The Classical Methodological System of Training-Data Influence Estimation: LOO, Shapley, and Influence Functions Training-data influence estimation rests on three paradigms. Leave-one-out (LOO) [47] removes a datum and retrains the model, providing the most direct causal measure but with cost linear in dataset size, making it infeasible for large-scale pre-training. Shapley value [48] allocates each datum’s marginal contribution across all subsets, but likewise requires expensive retraining. Influence functions [1] approximate LOO via first-order Taylor expansion and are more efficient; TRAK [108] extends this paradigm to ImageNet-scale models using random projection and linear output fitting.

These paradigms assume permutation invariance: learning outcomes do not depend on data order, allowing influence to be decomposed into individual contributions. Wang et al. (ICLR 2025) [49] challenge this assumption because stochastic, non-convex, multi-stage foundation-model training is order-sensitive. They formalize Trajectory-Specific LOO (TSLOO), measuring the effect of removing a datum at a specific training iteration along the exact optimization trajectory. TSLOO depends on timing and captures interactions: later similar data can attenuate earlier influence. They further show that influence functions approximate expected LOO only when checkpoints and Hessians are approximately identical and the learning rate is constant, conditions rarely satisfied in practice [49].

Systematic Failure of the Data-Independence Assumption: Non-Convergence and Sequential Arrival TSLOO can be efficiently approximated by Data Value Embedding (DVEmb), which extracts from its first-order Taylor expansion components independent of test data and represents a datum’s cumulative trajectory effect; influence is then obtained through a dot product with test-data gradients. On Pythia-410M [50], influence exhibits three phases: a short high-influence warm-up, a low-influence basin, and a later rise. Selecting only before 2,000 steps and after 20,000 steps achieves nearly full-horizon performance while reducing overhead by $> 5\times$; DVEmb uses 0.84 GB GPU memory versus 63.6 GB for LoGRA [51], improving throughput by $> 15\times$.

Nishida et al. [52] define training-sample influence without a downstream task or validation set as the reduction in squared L2 distance to the final parameters: $Cont(B_t) = 2\Delta_t^\top(\theta^* - \theta_t) - \|\Delta_t\|_2^2$. The first term measures alignment toward the final parameters and the second penalizes update magnitude. Across 18 Pythia [50] and PolyPythia [53] configurations, literature data align more strongly early while STEM data do so later. Adversarial samples increase from roughly 0–1% to 8–10% late in training, showing declining reliability of per-sample scores. Checkpoint approximations correlate strongly with exact values late (Spearman > 0.94) but only moderately early (≈ 0.6), reflecting larger parameter changes early in training.

SOURCE (Segmented StatiOnary UnRolling for Counterfactual Estimation, NeurIPS 2024) [54] directly addresses non-convergence by partitioning training into segments and treating gradient/Hessian distributions as stationary within each. It requires neither a unique optimum nor convergence and handles multi-stage pipelines such as continual learning [55] using only a few checkpoints (e.g., 6), outperforming existing attribution methods especially in non-converged and multi-stage settings. Sow et al. [56] instead dynamically reweight samples online according to loss, with a theoretical framework characterizing its effect on gradient-optimization convergence, providing a route to assessing sample value during training without changing the data.

Optimizer Dependence and the Breakdown of Influence Functions: From SGD to Adam

Classical influence functions [1] are derived from the SGD update rule [57], whose perturb–reweight–reoptimize analysis follows the raw gradient direction. Modern LLM pre-training instead relies heavily on AdamW [59], whose momentum and second-moment normalization alter the effective update direction. Thus, raw-gradient scoring can be geometrically inconsistent with actual training dynamics. Although TSLOO captures trajectory dependence, its efficient DVEmb approximation remains SGD-based and cannot be directly extended to Adam because of normalization terms [49]. This SGD-to-Adam gap provides the missing conditioning term in the error decomposition of RQ1.

TrackStar [60] scales gradient-based attribution to over 160B tokens for an 8B language model without

subsampling or pre-filtering, combining optimizer-state correction, task-specific Hessian approximation, and normalized encoding. Yet on fact-tracing tasks it is better at identifying samples influencing model predictions, whereas BM25 [61] better retrieves passages explicitly containing the relevant fact, revealing a gap between attribution and actual knowledge contribution. This suggests that proxy-value misalignment combines optimizer-geometry conditioning with limited proximal horizons, whose relative contributions motivate the RQ1 error decomposition.

Adam-specific attribution remains sparse. Bayesian Influence Functions [62] uses stochastic-gradient MCMC for Hessian-free attribution but still assumes SGD dynamics. In-Run Data Shapley [63] introduces dynamic valuation, while its Adam extension [64] finds SGD-based proxy estimates correlate with true contributions at only about 0.11; frozen-state and linearized Ghost approximations restore $R > 0.99$ while retaining about 95% throughput. OPUS [2] directly defines utility in optimizer-induced update space: for AdamW, $P_t^{AdamW} := C_t \cdot \text{Diag}(1/(\hat{v}_t + \epsilon))$, and projects a candidate’s preconditioned update onto a proxy direction. With Ghost [63] and CountSketch [65], it adds only 4.7% compute; on GPT-2 XL, 30B tokens surpass a full 200B-token run, while on Qwen3-8B continued pre-training, 0.5B tokens surpass 3B tokens. Nevertheless, OPUS and Adam-aware In-Run Data Shapley remain proximal and per-sample: neither combines Adam momentum with usage timing, nor predicts the total cost from the current state to a target capability. Sequential dependence is likewise not explicitly modeled.

5.2 Data-Refinement Methods for Datasets Yet to Be Formed

Domain-Level Data-Mixture Optimization for Datasets Yet to Be Formed As data-scoring methods fail under non-convergence and sequential arrival [49][54], research has shifted partly toward ex-ante refinement [8][9], especially domain-level mixture optimization [39], ranging from explicit mixture fitting and data experts [43] to tokenizer-regression estimation [98]. DoReMi [111] learns per-domain weights through group distributionally robust optimization on a small proxy model and transfers the mixture to large-scale pre-training. ADO [66] optimizes the data distribution online using domain-specific scaling laws, requiring neither external knowledge, proxy models, nor update modifications. Its scaling-law extrapolation also provides a basis for anticipating valuable domains at larger scales. Although ADO assumes sample exchangeability, its domain-level view bridges per-sample scoring and refinement.

DataDecide [67] provides the largest public suite for data decisions, training 14 model scales from 4M to 1B parameters on 25 sources, deduplication and filtering strategies, producing $> 30,000$ checkpoints. It asks how small experiments can predict the best corpus at scale: a single 150M-scale ranking predicts 1B decisions with about 80% accuracy, while scaling-law methods do not outperform this baseline; intermediate checkpoints perform similarly to compute-matched final checkpoints. Thus simple domain-level rankings already approach the prediction boundary.

Dynamic Gradient Alignment (DGA) [68] estimates pre-training mixtures online by aligning gradients on specific tasks with those on pre-training data. It outperforms importance sampling when the pre-training set is small or task-specific data is scarce, demonstrating a feasible route to dynamic valuation from training history.

These methods cover domain mixing [8][66][68], joint batch curation [31][32], difficulty-controlled synthesis [69], and curriculum learning [20], but retain sample exchangeability. ADO and DGA dynamically use scaling laws or gradients, while topic-aware reweighting [42] and bi-level selection-utilization optimization [23] follow the same direction. None models sequential dependence in trajectory data. DataDecide [67]

establishes an approximately 80% prediction floor using lightweight representations, while also suggesting limited headroom for scaling-law extrapolation. Efficient refinement therefore still requires data valuation.

Difficulty-Controllable Data Synthesis and the Curriculum-Learning Paradigm Data refinement also controls synthetic content and difficulty [10][28] and automatically constructs corpora [99]. MathMixup [69] generates high-quality, difficulty-controllable mathematical problems through mixing and decomposition, combining self-checking, manual screening, and curriculum learning. DiReCT [22] further shows that timing-aware selection during annealing improves model quality under equal token budgets, emphasizing both what data to introduce and when.

Thinking Augmented Pre-training (TPT) [70] augments existing text with reasoning traces because reasoning complexity makes some high-quality tokens difficult to learn at fixed capacity. Across configurations up to 100B tokens, TPT improves pre-training data efficiency by $3\times$; for a 3B model, downstream reasoning performance improves by $> 10\%$. It thus extends refinement from generating new data to increasing information density.

DeepSpeed Data Efficiency [71] combines curriculum-based sampling with hierarchical token dropping. For GPT-3 1.3B, it achieves $12.5\times$ savings in data/time/cost while retaining 95% quality (3.7K vs. 46.3K Azure); under matched quality, $2\times$ savings are also possible. This quantifies the potential efficiency gain from data organization.

Pre-training under Infinite Compute [72] studies fixed-data, unconstrained-compute training. Increasing epochs and parameters causes overfitting, mitigated by stronger regularization; the optimal weight decay is $30\times$ standard practice. The best interventions reach the same asymptote on 200M tokens using $5.17\times$ less data, and an $8\times$ smaller student retains 83% of the ensemble benefit. However, it addresses neither per-sample selection nor value prediction from model states.

Active Collection and Ex-Ante Refinement Strategies for Embodied-Intelligence Data

Embodied-intelligence data differs from NLP/CV because collection requires physical robot–environment interaction and is costly [13][46]. MimicGen [100] generates large-scale demonstrations from a small number of human demonstrations but performs no valuation. A survey in Data Acquisition and Processing emphasizes large-scale, high-quality VLA data and standardizes action alignment, multimodal synchronization, language normalization, and quality control.

Jing and Kong [83] propose EITS, a self-supervised exploration strategy for informative embodied-perception trajectories. A 3D semantic-distribution map and exploration policy use semantic divergence and uncertainty rewards to explore objects with uncertain or differing cross-view distributions; difficult samples are then selected along trajectories. This directly addresses which trajectories/samples to collect and provides a basis for extending data refinement from exchangeable samples to sequentially dependent trajectories.

5.3 Lightweight Prediction and Stopping-Criterion Computation from Model-Internal Representations and Optimizer State

Lightweight Data-Influence Prediction from Model Representations and Training Trajectories

MATES (Model-Aware Data Selection for Efficient Pretraining with Data Influence Models) [73] probes the current model with a single-step gradient update, measures loss change on a reference set, and

trains a small influence model to approximate this oracle signal using last-hidden-layer representations. The process repeats across stages as model states change. On 410M and 1B C4 [74] pre-training, MATES outperforms random selection, halves FLOPs to a target performance, and doubles the gains of DsDm [80]. Zero-Shot Data Maps [15] provide an even lighter diagnostic by predicting training-dynamics maps from a pre-trained model’s loss distribution without target-data training.

Group-MATES [75] extends individual influence to groups and shows that additive individual scoring fails: after selecting only 100 points, overlap with brute-force group selection approaches random levels, consistent with trajectory-dependent influence [49]. It uses relational weights $R_{x_i, x_t} = \text{sim}(h_{x_i}, h_{x_t})$ to model cancellation and amplification. On DCLM [76], it improves 3.5%–9.4% over random across 400M-4x, 1B-1x, and 3B-1x settings, nearly doubling individual-selection gains and reducing tokens to a target performance by up to $1.75\times$. Thus it explicitly challenges additivity, though its relations remain embedding-based rather than sequential.

BLISS [77] models long-term influence to convergence through a bi-level framework with a proxy model and scoring model. The upper level learns sample weights while the lower level trains the proxy to convergence; KL divergence [78] aligns the proxy with the LLM. On Pythia 410M/1B/2.8B and LLaMA-0.5B, BLISS achieves a $1.7\times$ speedup over MATES at 1B; selection costs only 10–20% of total FLOPs.

Approx-LESS [79] reduces gradient-based influence cost by computing LoRA gradient features [81] for a small subset and regressing influence for the remainder. On 270K instructions, it matches LESS [26] closely while reducing gradient-extraction time by $> 3\times$, demonstrating scalable influence prediction from lightweight model representations.

Optimizer-State-Aware Data Valuation OPUS provides the most systematic use of optimizer state for data valuation. It defines $U_z(t) \approx \eta_t \langle u_z(t), g_{\text{proxy}}(t) \rangle - \eta_t^2 \langle u_z(t), G(t) \rangle$, where $u_z(t) = P_t \nabla_{\theta} L(z; \theta_t)$ is the optimizer-induced update direction; the first term measures alignment and the second redundancy. Its AdamW preconditioner $P_t^{\text{AdamW}} := C_t \cdot \text{Diag}(1/(\hat{v}_{t-1} + \epsilon))$ incorporates the second moment directly. On FineWeb-Edu, OPUS outperforms the quality-4/5 baseline even when selecting quality-3 data; on Qwen3-8B continued pre-training, 0.5B tokens surpass the full 3B-token run, giving a $6\times$ efficiency gain.

Although Group-MATES lacks explicit optimizer state, its relational weight is theoretically linked to the trajectory-specific influence product $\prod_{j=t_s+1}^{T-1} (I - \eta_j H_j)$ [75]. Replacing the Hessian approximation with the optimizer preconditioner therefore suggests a route toward optimizer-aware relational influence.

QuaDMix [95] optimizes pre-training data distributions under fixed budgets by jointly modeling quality and diversity, using small-model simulation and LightGBM for parameter search. DATAMASK [96] treats selection as mask learning with policy gradients and selects about 10% of 15-trillion-token FineWeb, improving 3.2% on a 1.5B dense model and 1.9% on a 7B MoE. Neither uses remaining training budget as a dynamic decision input or predicts how compute changes after data introduction.

Remaining-Budget-Aware Joint Compute–Data Budgeting Joint data–compute optimization with remaining budget as an explicit input is the most challenging goal. Compute-Constrained Data Selection [84] first incorporates selection cost and training gain into a unified budget; its follow-up [103] further studies compute-constrained selection but remains text-only and fixed-budget. Many powerful methods, including gradient selection [26], are not compute-optimal: cheaper perplexity selection [24] can dominate, with trainer-to-scorer ratios of $5\times$ for perplexity and $10\times$ for gradients. However, the framework

targets fine-tuning rather than dynamic pre-training allocation.

Scaling Data-Constrained Language Models [85] treats compute C and unique-data budget D_C as explicit constraints for allocating parameters N and tokens D . It models effective data D' and finds that repeated-data value decays sharply after roughly 16 epochs; under data constraints, additional compute should favor epochs rather than parameters. It addresses data-parameter allocation but not per-sample selection or model-state-based valuation.

Functional Scaling Laws [86] derive loss-trajectory scaling laws under arbitrary learning-rate schedules, while Scaling with Collapse [87] shows that optimally trained LLM loss curves collapse onto a normalized universal trajectory. Deviations can diagnose training pathologies and potentially predict trajectory changes after new data, but this has not yet been used directly for data selection or compute prediction.

Overall, Compute-Constrained Data Selection [84] and Data-Constrained Scaling Laws [85] do not jointly predict, from model parameters, optimizer state, and remaining budget, how introducing a dataset changes global compute consumption without training. Three unsolved components remain: model-state-based value prediction [73][75][77], remaining-budget-conditioned selection [84], and prediction of post-introduction trajectory and compute changes [86][87]. More fundamentally, these methods retain sample exchangeability; Group-MATES models relations but not sequential dependence, motivating the trajectory-data problem addressed below.

5.4 Approaches to Characterizing Temporal Dependence in Embodied-Domain Trajectory Data

Sequential Characteristics of Embodied Trajectory Data Embodied trajectory data differs fundamentally from text [13][82]. Its meaningful unit is an interaction episode containing physical actions, multimodal observations, and outcomes. Physical coherence requires temporal synchronization and spatial alignment [88]. For camera-LiDAR detection, an IoU threshold of 0.5 tolerates 273 ms at 5 m/s but only 34 ms at 40 m/s; a 40 ms camera-IMU offset can cause up to 10 m translational and 3° rotational error; and an 849 ms inter-robot offset can change velocity estimation from 6.30 to 2.34 m/s. Humanoid datasets combine cameras, depth, IMU, joint states, contact, tactile, torque, object poses, and commands across coordinate frames, requiring calibrated, timestamped transformations [82]. These dependencies provide the physical basis of sample ordering.

The Comprehensive Breakdown of the Exchangeability Assumption on Trajectory Data, and Existing Responses Trajectory data violates sample exchangeability because observations are temporally dependent. Alvarez et al. [89] explicitly identify this issue for conformal prediction and propose methods based on trajectory regularity, high-frequency calibration, and random sampling-time assumptions. Time-series and sequential auto-transition data exhibit the same violation [90].

In offline reinforcement learning, Jia et al. (COLT 2024) [91] show that offline policy-evaluation sample complexity is governed by the concentrability of the aggregated Markov transition model, which can grow exponentially with horizon even when the original MDP is well behaved. Thus per-sample scoring errors may amplify with trajectory length. They also show, under value-function realizability, that hard instances can be reduced to trajectory-data settings, providing no inherent advantage for trajectory data. Farias et al. [92] study Markovian interference, where interventions on units affect others through dynamic constraints, providing a related formal view of dependence.

Several conformal-prediction works address non-exchangeable temporal or structured data [90]. Efficient Quantification of Time-Series Prediction Error [93] parameterizes nonconformity measures and optimizes them explicitly, while Structured Conformal Inference for Matrix Completion [94] develops weighted conformalization without exchangeability. Although these works concern uncertainty rather than data influence, their treatment of temporal and structured dependence provides methodological references for influence estimation on non-exchangeable trajectory data.

6 Synthesis: what the field can now do, and what remains unresolved

Research Strand	Representative methods	Capability Established	Unsolved Issue for this Proposal
Static scoring and retrospective valuation	FineWeb-Edu [24], DSIR [106], QuRating [107], Dataset Cartography [12], Zero-Shot Data Maps [15], Data Pruning [16], Diversity-Aware Selection [19], LOO [47], Shapley [48], Data Shapley [6], Datamodels [7], TRAK [108]	Static quality, diversity, and training-dynamics scores; counterfactual valuation via LOO, Shapley, and learned subset–performance mappings.	Static scores ignore training state; retrospective value does not quantify future savings.
Influence, proximal, and optimizer-aware scoring	Influence Functions [1], Group Effects [17], LoGRA [51], TrackStar [60], Bayesian IF [62], SOURCE [54], LESS [26], GIST [30], CHIPS [4], Approx-LESS [79], Rho-1 [3], NICE [21], GradAlign [110], OPUS [2], In-Run Shapley [63,64], DiReCT [22]	First-order and proximal scores estimate immediate or targeted utility; optimizer-aware methods incorporate AdamW or related update geometry.	Failure conditions, global horizon, and trajectory dependence remain incompletely modeled.
Temporal and lightweight influence estimation	TSLOO/DVEmb [49], Nishida [52], PolyPythias [53], Dynamic Loss Reweighting [56], MATES [73], Group-MATES [75], BLISS [77], DWM [23], JEST [31], SALN [32]	Training-history-aware influence, phase-dependent valuation, group interactions, lightweight proxy prediction, and bi-level long-term estimation.	No unified dataset-level counterfactual under a remaining budget.
Domain mixing, refinement, and synthesis	Data Mixing Laws [8], DoReMi [111], DoGE [9], ADO [66], DGA [68], DataDecide [67], QuaDMix [95], DATAMASK [96], Optimal Mixture Scaling [39], ToReMi [42], MiDE [123], TREX [98], Textbooks Are All You Need [28], MetaMath [10], MathMixup [69], TPT [70], AutoData [99]	Learns domain weights, quality–diversity trade-offs, and data-generation policies; synthesis improves data efficiency under controlled settings.	Remains largely domain-level and ex-ante; training history and per-unit value are not jointly modeled.
Curriculum and compute–data optimization	Curriculum Learning [20], Iterative Machine Teaching [5], Easy-to-Hard [11], DeepSpeed Data Efficiency [71], Chinchilla [25], Data-Constrained Scaling [85], Compute-Constrained Data Selection [84], Compute-Optimal Beyond Fixed Durations [40], Functional Scaling Laws [86], Scaling with Collapse [87], Infinite Compute [72]	Establishes order effects, compute–data trade-offs, optimal stopping, and data-efficiency scaling under fixed or constrained budgets.	Dataset-level cost savings and adaptive selection are not jointly optimized.

Research Strand	Representative methods	Capability Established	Unsolved Issue for this Proposal
Embodied data scaling and curation	OXE [13], DROID [34], Data Scaling Laws [35], What Matters [46], Re-Mix [36], Robot Utility Models [127], Curse of Precision [101], Robot Scaling Laws [102], MimicGen [100], EITS [83], Active Learning [29], Dataset Distillation [27]	Characterizes dataset mixtures, scaling laws, informative trajectory collection, uncertainty-driven annotation, and data synthesis for embodied learning.	Operates mainly at dataset or mixture level; no unified trajectory-level value function.
Trajectory influence and non-exchangeability	CUPID [37], ATHENA [38], Demo Curation via IF [113], SCIZOR [114], Action Shapley [100], Decoupled Prioritized Resampling [101], Causal Confusion in IL [115], Conformal Trajectories [89], Beyond Exchangeability [90], Jia COLT [91], Markovian Interference [92], Tkachuk [97], OSCP [93], Structured Conformal [94]	Establishes trajectory-level curation, action influence, temporal dependence, Markovian interference, and statistical tools for non-exchangeable data.	Cross-segment causal coupling is largely unmodeled and remains disconnected from data valuation.
Audits, benchmarks, and field synthesis	Albalak Survey [116], Data-centric AI Survey [41], Analyzing Similarity Metrics [117], Critical Look at Targeted Selection [118], DataComp-LM [76]	Provides taxonomies, large-scale benchmarks, reliability audits, and evidence on representation–algorithm interactions.	Granularity, training drift, and distribution shift remain fragmented; optimizer state and trajectory dependence are not unified.

Summary of Existing Research Gaps

First, the sample-exchangeability assumption fails systematically under modern training paradigms: the TSLOO framework proves that influence functions approximate expected LOO scores only under oversimplified conditions—all checkpoint parameters approximately identical, all Hessian matrices approximately identical, constant learning rate—conditions that almost never hold in LLM pre-training [49]. The temporal dependence of data influence induced by non-convergence and sequential arrival has been empirically quantified—data influence exhibits a three-phase pattern (high-influence warm-up, low-influence basin, gradual rise), with checkpoint approximations only moderately correlated in early training (< 0.6) and highly consistent later (> 0.94) [49].

Second, data-refinement and lightweight-prediction methods have partially achieved the goal of estimating data value without actual training: MATES predicts influence via linear regression on last-hidden-layer representations, halving FLOPs [73]; Group-MATES captures inter-point interactions via relational weights, reducing tokens by $1.75\times$ [75]; BLISS explicitly models long-term influence via bi-level optimization, attaining a $1.7\times$ speedup [77]; OPUS incorporates optimizer-internal state into utility computation, achieving a $6\times$ data-efficiency gain [2]. Third, joint compute–data budget optimization has preliminary frameworks: Compute-Constrained Data Selection first brings selection cost into the total budget [84]; Data-Constrained Scaling Laws treats compute and unique-data budgets as explicit constraints [85]; the

loss-curve collapse phenomenon makes training-trajectory prediction plausible [87]. Fourth, the causal-dependence character of trajectory data has been characterized along multiple dimensions—quantified physical-coherence requirements, statistical violation of exchangeability, and the theoretical finding that concentrability grows exponentially with trajectory length [91]—but no unified framework yet incorporates causal dependence into data-influence estimation.

A possible path to unification starts from the TSLOO theoretical framework. TSLOO’s cumulative-effect term $\prod_{k=t_s+1}^{T-1} (I - \eta_k H_k)$ can, conceptually, be generalized to trajectory data: removing any sample within a trajectory changes the distribution of subsequent states, and this causal propagation is precisely what the cumulative product captures. Further replacing the Hessian approximation with an Adam optimizer preconditioner (as OPUS does) would let the same framework handle optimizer state and trajectory dependence together. Group-MATES’s relational weights offer another path: by explicitly modeling relations between data points to transcend the sum of individual influences, such relations extend naturally to causal dependencies. The key challenge is achieving efficient computation under causal dependence—DVEmb’s gradient decomposition, random projection, and influence-checkpoint techniques provide the engineering basis for scalability.

7. Research Questions and Hypotheses

RQ1: Systematic Characterization of Failure Boundaries

RQ1.1 Predictability of Failure Boundaries When the object of evaluation changes from a single identically distributed sample to an entire dataset distributionally shifted from the data already seen, how does the deviation between the scores of static scoring [24][106][107] and proximal targeted scoring [3][26][30] and the global expected gain vary jointly under the threefold deviation—distribution-shift magnitude, evaluation granularity (sample-level $\rightarrow$ dataset-level), and optimizer-geometry conditions (SGD assumption $\rightarrow$ actual AdamW/Muon dynamics [2][64])?

H1.1 The deviation exhibits a threshold structure rather than gradual degradation—proximal scores remain usable below a threshold shift distance and fail rapidly beyond it—and the threshold location can be predicted by precursor quantities extractable from the training history (gradient-alignment decay [4][110], curvature-spectrum shift [4], optimizer second-moment drift [2]). The predictive validity of these precursors is assessed against two anchors—the ex-post true failure tier as the upper-bound reference and random prediction as the lower-bound reference—with the specific criterion calibrated on Failure-Atlas and pre-registered.

RQ1.2 Sources of Error under Sequential Drift When data value decays or migrates as training proceeds (diminishing returns on repeated reuse [85]; order sensitivity [11][49]), where does the scoring error of methods with proximal-improvement utility come from, and what share does each source account for?

H1.2 The error admits a four-term decomposition—truncated horizon, missing conditioning, broken exchangeability, and selection-feedback shift—that explains it nearly exhaustively. Each term’s expected share is regime-specific: the truncated-horizon term dominates in long-horizon tasks, the

broken-exchangeability term in shift settings, and the selection-feedback term in recursive self-training settings (model collapse [44] as its extreme form).

RQ1.3 Well-Definedness of Independent Scoring When temporal dependence exists between samples (trajectory data [13][89][91]) or the act of selection itself alters the distribution of subsequent data [44][49], is "per-sample independent scoring" still well-defined mathematically?

H1.3 It is not—and the degree of ill-definedness is measurable: sample value contains non-negligible interaction terms (value coupling between segments of the same trajectory; changes that already-selected data induce in the value of unselected data), whose strength grows monotonically with temporal coupling and selection pressure. When the interaction-to-main-effect ratio exceeds a threshold, the ranking produced by independent scoring departs significantly from the true-value ranking—in high-coupling strata, the rank correlation of segment-wise evaluation [37] falls below the random baseline, while trajectory-level joint evaluation remains significant.

RQ2: The Prospective Value Function and the Two Tools

RQ2.1 Estimability of Prospective Value Can the value function

$$V(D \mid H_t, B) = \mathbb{E}_A [J(\theta_T^D)] - \mathbb{E}_A [J(\theta_T^{\mathcal{D}})]$$

be estimated to decision-grade precision under a zero-training-cost constraint?

H2.1 Yes—conditioned on a summary of the training history, the estimator follows a "backbone plus corrections" construction: curvature-aware influence [4][17] as the backbone, the optimizer-induced update space [2][64] as the geometry, and four correction modules—one per error source isolated in RQ1.2—composed in the canonical causal order (geometry conditioning first, then group aggregation, then horizon extrapolation, feedback tracking last) into $\mathbf{V}$. Its scores correlate with true training gains significantly more strongly than both baseline classes—static scoring [24][106][107] and proximal targeted scoring [26][30]—with training-requiring retrospective valuation [6][7] as the precision upper-bound reference; selection overhead remains a small fraction of total training cost; and, under the condition that the proxy aligns with the true optimization geometry, a provable lower bound on the correlation exists (extending the CHIPS proof paradigm [4] from the single-shot, image-text-pair setting to the sequential, dataset-level one).

RQ2.2 The Gain Mechanism of Targeted Production For data yet to be formed, can targeted synthesis and collection guided by the value function save more of the total cost to reach a target capability than equal-budget non-targeted production?

H2.2 Yes, and the gain mechanism is localizable—targeted returns concentrate in the "capability boundary band" (the gap region between the current state and the target, as marked by the estimator); at equal target capability, targeted production significantly reduces token consumption relative to random synthesis [28] and natural collection (threshold to be calibrated on Failure-Atlas). Falsification design: if targeting is replaced by "random synthesis within the boundary band," the gain should attenuate significantly; if it

does not, the mechanism narrative narrows to the combination of "region selection + random production" (predefined downgrade path).

Bridge from RQ1 to RQ2 the diagnosis is the prescription. RQ1 delivers an error model; RQ2 is required to build its estimator out of that model. This constraint distinguishes the present construction from all prior valuation tools: every component of V represents a measured error source that demands it, and the two claims are jointly testable.

RQ3: Generalization to Trajectories and Sequential Decision-Making

RQ3.1 The Value Missed by Temporal Structure Within the cross-segment temporal structure of trajectory data, how large is the value interaction missed by segment-wise independent evaluation [37][38], and how much of it can temporally aware scoring recover?

H3.1 The interaction terms missed by segment-wise evaluation (as measured under RQ1.3) are non-negligible on real trajectory data and grow with the task's temporal-coupling strength; a scorer with the three-layer structure—segment-level temporal correction, trajectory-level group estimation [17], and distribution-level shift gating [4]—stably outperforms segment-wise baselines [37][38] in cross-embodiment mixed training [13], with gains stratified along the coupling-strength ladder of the evaluation suites (Long > Goal > Spatial/Object)—the stratification pattern itself is the core falsifiable prediction. (Verification: D9 TrajScorer / E3.1)

RQ3.2 Trackability of the Stopping Criterion In trajectory-data training, can the marginal gain of continued training be tracked in a training-free manner, thereby yielding a stopping criterion?

H3.2 Yes—the marginal-gain sequence can be tracked by the estimator with bounded error (median relative error ~ 15%); the stopping point derived from "marginal expected gain - marginal cost" wastes at most 10% of total cost relative to the ex-post optimal stopping point (oracle), significantly outperforming both prevailing practices—fixed-budget stopping and loss-plateau criteria—and aligning with the predictive form of text-side compute-optimal theory [40]. (Verification: D11 SeqDecider / E3.2)

RQ3.3 The Source of Cross-Modal Generalization Is the cross-modal generalization of the value function rooted in a shared conditioning structure or in modality-specific fitting—can one and the same value function cover both exchangeable samples and sequentially dependent trajectories with zero tuning?

H3.3 The former—when the core components of the text-side estimator (history summarizer, curvature-aware backbone [4], optimizer geometry [2]) are deployed on trajectory data with all parameters and hyperparameters fully frozen, changing only the data interface and modality-specific preprocessing, it still significantly outperforms both the history-free baseline and the strongest trajectory-side specialized baseline [37]; ablations show that the history-conditioning components (model-parameter summary / optimizer state / seen-data summary), rather than modality terms, are the largest source of contribution. Should generalization vanish, the unification claim weakens into a design principle plus independent validation on both modalities—the negative result would itself clarify the boundary of "unification" and likewise constitutes a publishable conclusion (pre-specified exit). (Verification: D12 UniBench / E3.3)

8. Experimental Analysis and Falsification Tests

E1.1 → H1.1: Shift-granularity grid experiment. Design. The experiment uses continued pre-training of a 1B model as its vehicle. The model is first trained on the source domain (a C4 subset) up to a designated checkpoint; we then construct a sequence of candidate datasets with monotonically increasing distributional shift: same-domain resampling → intra-domain subdomain → neighboring domain → distant domain (shift magnitude quantified into five tiers via the dual metrics of embedding-space distance [117] and MMD). The methods under evaluation—OPUS [2], Rho-1 [3], LESS [26], and the quality classifier [24]—score within each tier at both sample-level and dataset-level granularity, with small-scale ground-truth training gains serving as the oracle. Measurement. Spearman rank correlation between each method’s scores and the oracle gains, plotted against shift magnitude; in addition, threshold-prediction regressions on precursor quantities drawn from the training history (decay of gradient alignment, shift of the curvature spectrum). Expected outcome. The rank-correlation curve exhibits a "plateau-cliff" threshold structure rather than uniform degradation. Falsification criterion. If the curve declines uniformly or the precursors show no predictive power, the threshold structure of H1.1 fails to hold, and the map is re-characterized in terms of "conservative usage intervals" (corresponding to the pre-specified fallback path).

E1.2 → H1.2: Four-source error decomposition experiment. Design. The core idea is modular completion: for each proximal scoring method, we successively "supply" its missing component and observe the resulting error reduction, which directly quantifies that error source’s share. Each completion is realized as a deliverable correction module destined for $\mathbf{V}$ —truncated horizon → piecewise-rollout extrapolation along the remaining training trajectory (horizon-extrapolation module, SOURCE lineage [54]); missing conditioning → utility under the optimizer’s actual geometry (geometry-conditioning module, OPUS/Pion lineage [2], [105]); exchangeability breakdown → relation-weighted group-level aggregation (group-aggregation module, Group-MATES lineage [75]); selection feedback → a distribution-tracking term over the shift induced by selection (distribution-tracking module). Measurement. For each module: (i) the gross reduction when applied in isolation; (ii) the net reduction under cumulative composition in the canonical causal order; (iii) the residual interaction term (total error minus the sum of the four reductions). The difference between (i) and (ii) constitutes a direct measurement of inter-module interaction. Expected outcome. The four terms jointly explain the error variance; each module’s net reduction remains within the calibrated margin of its gross reduction; the ranking of shares varies across settings in a manner consistent with the formulation of H1.2. Falsification criterion. If the joint explanatory power falls below the pre-registered criterion, we revert to qualitative characterization.

E1.3 → H1.3: Interaction-term nested-measurement experiment. Design. On trajectory data, three evaluation modes are constructed: (a) independent segment-wise evaluation (CUPID-style [37]); (b) trajectory-level joint evaluation; (c) the oracle (ground-truth training). Interaction-term strength is defined as (b)’s estimate minus the segment-wise sum from (a)—that is, the cross-segment coupling value that segmentation omits. Sampling is stratified by the tasks’ temporal-causal coupling strength (the stratification protocol follows the task-dimension analysis of large-scale robot datasets [46]: pick-and-place tasks as low coupling, long-horizon multi-stage tasks as high coupling), with the interaction-to-main-effect ratio measured independently within each stratum. Measurement. The curve of the interaction/main-effect ratio against coupling strength; the rank correlations of (a) and (b) with (c), respectively. Expected

outcome. The ratio rises monotonically with coupling strength; in the high-coupling stratum, (a)’s rank correlation falls below the random baseline while (b) remains significant—directly verifying that ”independent scoring is ill-defined on highly coupled data.” Falsification criterion. If the interaction term remains below 10% of the main effect in every stratum, the segment-wise approximation is adopted and the center of gravity of RQ3 shifts toward the decision layer (pre-specified downgrade).

E2.1 → H2.1: Precision–cost bracketing experiment. Design. A three-tier bracketing structure—lower bound: static scoring [24]; middle tier: proximal targeted scoring [26], [30]; upper-bound reference: training-based Data Shapley [6] / Datamodels [7] (affordable only at the 150M tier); under test: our estimator $\mathbf{V}$; external criterion: ground-truth training gains. Scale ladder: 150M → 1B → 7B across three tiers, verifying that neither cost nor precision degrades with scale. Measurement. Correlation with ground-truth gains (paired tests against both baseline classes); the fraction of total training compute consumed by selection; LiSSA iteration convergence curves. Expected outcome. Correlation significantly exceeds both the static and proximal baseline classes; under proxy-alignment conditions the correlation lower bound is given by Theorem D4, and the empirical value should be no lower than the observable conversion of the theoretical bound. Falsification criterion. If no significant difference from the proximal baseline obtains, a component audit is initiated—ablating the information content of each HtHt component one by one, shrinking to the minimal effective set, and re-testing (as prescribed by the fallback path).

E2.2 → H2.2: Targeted-synthesis mechanism-localization experiment. Design. Mathematical-reasoning synthesis (MetaMath-style [10]) serves as the primary arena. The capability frontier band is defined by the estimator: difficulty intervals where the current model’s error rate is high yet influence-function values are high. Four arms: (i) targeted synthesis with our method; (ii) equal-budget random synthesis [28]; (iii) uniform synthesis across the full difficulty range; (iv) random synthesis within the frontier band—(iv) is the decisive control for mechanism falsification: if (i) and (iv) show no significant difference, the gain derives solely from ”selecting the right region” rather than targeting per se, and the mechanism narrative must be narrowed accordingly. Measurement. Token consumption for each arm to reach the same target performance; a decomposition map of gains by difficulty band (which intervals’ samples contributed the performance improvement). Expected outcome. (i) saves 30% tokens relative to (ii); the gain decomposition concentrates within the frontier band; (i) outperforms (iv) by at least one significance level. Falsification criterion. If (i) (iv), the contribution narrows to a combination of ”region selection + random generation”—still publishable, but with repositioned claims (corresponding to the fallback path).

E3.1 → H3.1: Coupling-stratified experiment for temporally aware scoring. Design. OpenVLA [33] as the backbone, OXE/DROID subsets [13], [34] as the data pool, and LIBERO [45] as the downstream evaluation arena—its four suites naturally form a ladder of temporal-coupling strength (Spatial/Object as short-horizon, low coupling; Goal intermediate; Long as long-horizon, high coupling). This stratification renders ”gains should increase with coupling strength” a falsifiable prediction rather than a post-hoc explanation. Three comparison arms: CUPID segment-wise scoring [37]; our three-tier structure (segment-level temporal correction / trajectory-level group aggregation / shift gating [17], [4]); random selection. Measurement. Per-suite success-rate improvement and training-token savings; a cross-suite trend test of gain versus coupling strength. Expected outcome. The largest advantage on the Long suite (significant

in both success-rate improvement and token savings), and parity with CUPID on Spatial/Object—the stratification of gains by coupling strength is itself the core prediction of H3.1. Falsification criterion. If the four suites show no gradient in gains, the interaction-term contribution is cast into doubt, and the E1.3 measurements are re-examined (the two experiments serve as mutual cross-checks).

E3.2 → H3.2: Stopping-criterion tracking experiment. Design. Two phases. Tracking phase: throughout training, at checkpoint intervals, the estimator outputs a "marginal-gain sequence," compared pointwise with the oracle marginal sequence (reconstructed post hoc from ground-truth training checkpoints), yielding the tracking error. Decision phase: a three-way contest among stopping strategies—fixed-budget stopping (the current default), the loss-plateau criterion (a common heuristic), and our gain-threshold stopping (stop once marginal expected gain $\geq$ marginal cost [40])—each assessed by total cost waste relative to the oracle-optimal stopping point. Measurement. The curve of tracking error (median relative error) over training progress; the cost-waste distributions of the three strategies (multiple random seeds). Expected outcome. Tracking error remains bounded, with a small median; the cost waste at our stopping point is significantly smaller than that of the fixed budget (fixed budgets waste more in settings where data value decays early); the stopping-point distribution aligns with the predicted form of text-side compute-optimal theory [40]. Falsification criterion. If tracking error is unbounded, the stopping criterion degenerates into upper-confidence-bound stopping (as prescribed by the fallback path).

E3.3 → H3.3: Zero-tuning frozen-transfer experiment. Design. Core-component freezing protocol—the estimator core developed on the text side (the history summarizer and the curvature-aware substrate) has all parameters and hyperparameters completely frozen; only the data interface and modality-specific preprocessing are replaced, and no quantity is re-fit. Three comparisons: direct transfer of the frozen components (under test); a history-free trajectory-side baseline; trajectory-side CUPID [37] (the strongest dedicated baseline). Additional ablation: HtHt components are switched off one at a time (removing the model-parameter summary / the optimizer state / the seen-data summary), with the corresponding trajectory-side performance drop observed—this answers whether generalization originates in the conditioning structure or in modality-specific fitting. Measurement. Score–effect correlation on both modalities; the component ranking of ablation-induced drops. Expected outcome. The frozen components significantly outperform the history-free baseline on the trajectory side; ablations reveal the history-conditioning components (rather than modality terms) as the largest source of contribution. Falsification criterion. If post-transfer performance falls to the level of the history-free baseline, the unification claim weakens into a design principle plus independent dual-modality validation, with the negative result itself constituting the contribution (pre-specified exit).

9. Research Plan

Phase 1 (Month 1–10) | Failure-Boundary Characterization

- **Month1–4:** Unified evaluation framework, UniVal-Framework-Lite

3 proximal-utility baselines under unified notation (OPUS / LESS / quality classifier); twofold-deviation protocol (dataset granularity $\times$ four shift tiers); unified error metrics. The sequential-drift protocol moves to Phase 4, sharing checkpoint infrastructure with the stopping criterion.

- Month5–7: Failure-boundary benchmark, Failure-Atlas-Lite

stratified grid slices over twofold deviation $\times$ magnitude $\times$ error measurement (retaining boundary-face grid points for threshold-structure tests); error-surface atlases for the 3 baselines; E1.1’s plateau-cliff curves and precursor threshold regression completed on the slices.

- Month8–10: Error-decomposition method, BiasDecomp + Paper 1

the four-term decomposition complete (truncated horizon / missing conditioning / broken exchangeability / selection-feedback shift); 3 correction modules fully delivered with independent measurement protocols, the selection-feedback term measured via controlled simulation; E1.2’s gross/net reduction and residual-interaction protocols unchanged. The group-aggregation module is shared with TrajScorer’s middle layer—built once, reused twice.

Deliver: UniVal-Framework-Lite + Failure-Atlas-Lite + BiasDecomp (3 modules + 1 simulated term) + Paper 1 finalized

Phase 2 (Month 11–20) | Value Function and Static Validation

- Month11–13: Prospective value function V-Value and guarantees

value-function definition unchanged (history conditioning, counterfactual baseline, expected gain); estimator $\bar{V}$ = proximal backbone + 3 correction modules (inherited from BiasDecomp); theorem system narrowed—the composition theorem (correlation lower bound for the correction family, bounded residual interaction; extending CHIPS’s paradigm from a single hybrid correction to a family) fully proved; of per-module guarantees, horizon extrapolation and geometry conditioning as full theorems, group aggregation and selection feedback as propositions + numerical support; still the draft of the thesis theory chapter.

- Month14–15: Estimator toolkit, EstV-Kit-Lite

training-history summary and Hessian–vector-product (LiSSA) modules; random projection and sketching via off-the-shelf implementations.

- Month16–18: Text-side static validation + targeted-synthesis validation

scale ladder 150M $\rightarrow$ 1B (7B as extended validation); one representative per layer of the squeezing comparison (static [24] / LESS [26] / Data Shapley [6]@150M); E2.1/E2.2 measurement protocols unchanged; E2.2 completed on the mathematical-reasoning domain alone—DataForge downgraded to single-round synthesis, the closed loop listed as future work.

- Month19–20: Paper 2 finalized

Deliver: V-Value definition + composition theorem + EstV-Kit-Lite + static validation results + targeted-synthesis validation + Paper 2 finalized

Phase 3 (Month 21–28) | Generalization to Trajectory Data

- Month21–23: Exchangeability-breakdown measure + TrajPipe-Lite

measure unchanged; OXE only (DROID cut); OpenVLA fine-tuning via existing implementations.

- Month24–26: Trajectory scorer, TrajScorer + E1.3 folded in

three-layer structure complete—segment-level causal correction, trajectory-level group estimation (reusing the Phase 1 group-aggregation module), distribution-level shift gating; vs CUPID + random selection; LIBERO’s three suites form the low/medium/high coupling ladder (Spatial / Goal / Long; Object folded into low coupling). E1.3’s nested interaction measurement reuses this pipeline; the E1.3–E3.1 mutual cross-check design retained.

- Month27–28: Frozen transfer + cross-embodiment demo + Paper 3

E3.3’s zero-tuning frozen-transfer protocol executed in full (the key evidence for the unification claim—not cut); X-Embodied-Select shrunk to a two-embodiment demonstration section within Paper 3 (both from OXE; “admission by value vs. by availability” comparison retained).

Deliver: measure + TrajPipe-Lite + TrajScorer + frozen-transfer results + cross-embodiment demo + Paper 3 finalized

Phase 4 (Month 29–40) | Decision Layer and Dissertation

- Month29–31: Streamlined stopping-criterion validation

three-variable decision narrowed to two (selection + stopping; mixture proportions via fixed grid sweep); sequential-drift measurement moved in from Phase 1; E3.2 two-stage protocol—tracking on the 1B text side, the three-policy stopping contest reusing trajectory-side checkpoints, no new full training runs.

- Month32–39: Unified evaluation protocol + dissertation

UniBench consolidated into a unified evaluation protocol (one value function, zero modality tuning, bimodal protocol and results tables reusing Phase 2/3 tasks and baselines—no new task suite, no leaderboard); decision-layer results as an optional workshop short paper or thesis chapters; dissertation writing and defense preparation.

Deliver: two-variable decision validation + unified evaluation protocol + workshop short paper (optional) + PhD dissertation

Four years’ timeline

Year	Phase	Research Question	Delivers
Year 1	Phase 1 Failure-Boundary Characterization	RQ1	UniVal-Framework-Lite, the unified evaluation framework; Failure-Atlas-Lite, the failure-boundary benchmark and error-surface atlases; BiasDecomp, the four-term error-decomposition method; Paper 1 finalized and submitted; Material for the doctoral thesis’s experimental chapters (failure boundaries and error decomposition); V-Value, the prospective value-function definition + composition-theorem proof initiated
Year 2	Phase 2 (first half) Value-Function and Tool Construction	RQ2	V-Value theorem system completed; EstV-Kit-Lite, the estimator toolkit; Text-side static validation results; Targeted-synthesis validation; Paper 2 released (ICML/ICLR); Trajectory exchangeability-breakdown measure + TrajPipe-Lite; TrajScorer, the trajectory scorer
Year 3	Phase 2 wrap-up → Phase 3 Generalization to Trajectory Data	RQ2→RQ3	TrajScorer completed; Frozen-transfer results + cross-embodiment mixed-training demonstration; Paper 3 released (CoRL/ICRA or NeurIPS); Two-variable empirical validation of the stopping criterion; Unified evaluation protocol
Year 4	Phase 3 wrap-up → Phase 4 The Decision Layer and Consolidation	RQ3	Unified evaluation protocol completed; Decision-layer results presented (workshop short paper, optional / thesis chapters) + doctoral dissertation finalized, submitted, and defended

10. Expected Contributions

1. A failure-boundary atlas for proximal scoring. For the online-selection utility family exemplified by [2][3][63], we will deliver the first systematic error characterization under the threefold deviation of distribution shift, evaluation granularity, and optimizer geometry—demarcating the trustworthy operating region for all subsequent training-free valuation work, and advancing [49]’s conditional characterization of when such tools fail into a decomposition of failure sources (where the error arises, and in what proportions)—with each error source characterized alongside a committable correction, thereby laying the groundwork for the construction of more complete data-refinement and data-scoring tools.
2. A dataset-level prospective value function and its training-free estimator, in place of next-step value. We define $V(D|H_t, B)$ —history-conditioned, with a counterfactual baseline and expected-gain semantics—and construct an estimator $\hat{V}$ carrying provable correlation lower bounds, generalizing [4]’s single-shot, image-text-pair setting to the sequential, dataset-level setting, and connecting theoretically to the counterfactual semantics of [1][17].
3. A sequence-structure-aware trajectory scorer. The three-layer design—segment-level causal correction, trajectory-level group estimation, and distribution-level shift gating—corrects the structural deficiency of segment-wise independent evaluation [37][38], while [54]’s training-dynamics perspec-

tive (segmented stationary unrolling) supplies complementary diagnostic signals on when within training the scorer’s assumptions hold.

4. A stopping criterion for the trajectory-data side. The optimal stopping condition ”marginal expected gain – marginal cost” is generalized from the text side [40] to trajectory-data training; together with mixture selection informed by the failure lessons of [36], it completes the decision layer of whether to select, how much, and how long to train—the three decisions a data-driven training pipeline must make.
5. A cross-modal validation system for unification. One and the same value function covers both text and trajectory data with no modality-specific tuning (with [4] and [37] as the respective modality-side baselines), delivered together with a reproducible evaluation framework—turning ”bridging the data barrier between LLM pre-training and embodied training” into a testable claim rather than a slogan.

11. Compute Budget Estimation

Principle: training-heavy labels and valuation-light computation. The compute expenditure of this project concentrates on two precisely budgetable batch workloads —small-scale oracle training runs that produce ground truth, and standardized continued-pretraining / fine-tuning pipelines. The valuation computation itself (V, TrajScorer, and all influence quantities) is training-free: LiSSA Hessian–vector products, random projection and sketching, and Ghost + CountSketch (the OPUS route’s measured overhead of $\sim 4.7\%$). All components of the research plan have likewise been compute-slimmed under the ”-Lite” principle (UniVal-Framework-Lite, Failure-Atlas-Lite, EstV-Kit-Lite, TrajPipe-Lite).

Phase-wise estimate (in A100 · hours, inclusive of multi-seed runs and repetitions; A6000/L40-class nodes converted at ~ 0.5):

Phase	Principal compute items	Estimate
Phase 1 (M1–10)	1B continued-pretraining baselines $\times 3$ (C4 subset); ~ 300 oracle ground-truth training runs (150M–400M, five shift tiers $\times$ dual granularity); scoring computation	3,000
Phase 2 (M11–20)	E2.1 precision–cost bracketing (Data Shapley upper bound at the 150M tier, main validation at 1B, 7B as extension only); E2.2 four-arm synthesis $\times$ multiple seeds; estimator development	5,000
Phase 3 (M21–28)	OpenVLA (7B) LoRA fine-tuning sweeps (3–4 coupling suites $\times 3$ comparison arms $\times 2$ –3 seeds, ~ 150 –200 hours per run); E1.3 reuses the same pipeline; E3.3 frozen transfer and the two-embodiment demonstration	7,000 (peak)
Phase 4 (M29–40)	Tracking and the stopping-policy contest fully reuse existing checkpoints; only new inference and lightweight analysis	1,000

1. Total and peak. Core experiments total $\sim 16\text{K}$ A100 · hours —an average sustained occupancy of roughly half an A100 over 40 months; with debugging, failed reruns, and queuing redundancy (engineering factor $\times 2$), $\sim 30\text{K}$ hours, i.e., an average sustained occupancy of about one A100. Peak

demand falls in Phase 3’s VLA fine-tuning sweep —a single 8-GPU node continuously occupied for 3–5 weeks, a job profile routinely absorbable by a shared cluster.

2. Storage and intermediates. The OXE shard subset, C4 / FineWeb-Edu subsets, and checkpoints (a few hundred MB per model under DVEmb-style compression) total ~50 TB, within a cluster’s routine storage quota; gradient-type intermediates are persisted exclusively in sketch / projected form, eliminating at the source the 63.6 GB-scale overhead of raw high-dimensional gradients.

12. Risks and Mitigation

This section focuses on the single risk of this proposal: the executability of the RQ3 robot-policy validation.

1. Risk identification. Physical robots or a maintained robotics simulation stack are probably not accessible, whereas the trajectory-side experiments (E3.1/E3.2/E3.3) superficially require “training policies and validating closed-loop performance.” Were this component non-executable, RQ3’s empirical evidence chain would lose its footing. This research has therefore structurally pre-empted the risk at the design stage and pre-positioned a four-level fallback.
2. Structural mitigation: the design consumes no physical platform. The embodied experiments are deliberately constrained to pure computation: the objects scored are public trajectory data (OXE), and the estimator consumes only data and model-internal quantities, never a physical system; policy training follows OpenVLA’s open-source implementation —in essence a standard GPU fine-tuning task; closed-loop evaluation is performed on LIBERO, a standard simulation benchmark built on robosuite/MuJoCo and run headlessly on compute nodes —amounting to “deploying a software stack” rather than “building a platform”; the sole cross-embodiment component (X-Embodied-Select) has been reduced to a data-level demonstration across two OXE embodiments, with no deployment involved.
3. Four-level fallback plan:
 - L1 : Switch the evaluation protocol. If simulation-stack deployment is blocked, the closed-loop success rate is replaced by an offline evaluation protocol —action-prediction error, held-out transition likelihood, and offline RL metrics. The core propositions of H3.1 (interaction terms non-negligible; independent scoring failing in high-coupling strata) and the conditioning-structure source of H3.3 are essentially data-level propositions, testable without closed-loop metrics; the rank correlation between offline proxies and closed-loop success rates can first be calibrated at small scale, thereby grading the evidentiary level of the offline protocol.
 - L2 : Reposition the claims. If closed-loop validation is wholly infeasible, RQ3’s empirical claims contract from “policy success-rate improvement” to “characterization of value structure on sequential data,” and Paper 3 moves from CoRL/ICRA to a NeurIPS/ICLR data track.
 - L3 : Collaborative supplementation. Partner, as a collaborator, with teams possessing real robots or simulation platforms to supply one demonstrative experiment. The need arises at Months 21–28 at the earliest, and the demonstration’s scale is low-burden for the partner.
 - L4 : Retract the center of gravity. If L1–L3 all prove infeasible, the dissertation’s weight retracts to the text side and theory —all claims of RQ1 and RQ2 stand independently on

the text modality, the first two papers remain unaffected, and the trajectory component is downgraded to a generalization chapter.

4. Decision mechanism. A go/no-go checkpoint is set before Phase 3 launches: an end-to-end feasibility run of LIBERO + OpenVLA fine-tuning on the cluster (single-run scale, ~ 200 A100 · hours). If it passes, the plan proceeds as scheduled; if blocked, the L1 offline protocol is feasibility-assessed within two weeks; if still infeasible, L2/L3 are triggered.

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